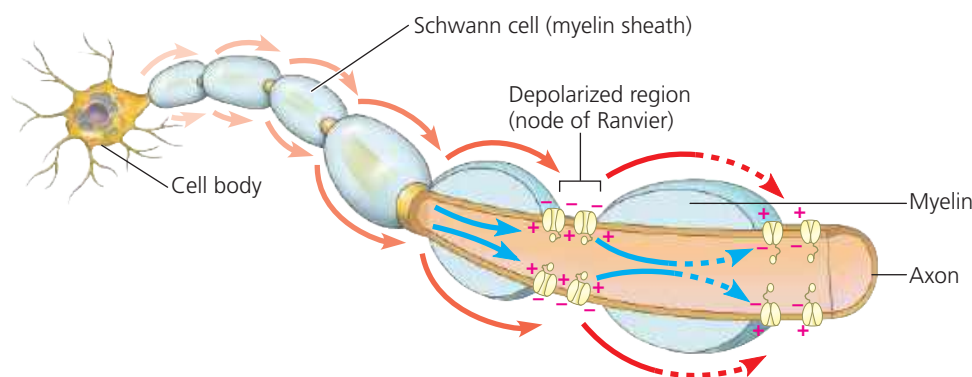


▼ **Figure 48.15 Propagation of action potentials in myelinated axons.** In a myelinated axon, the depolarizing current during an action potential at one node of Ranvier spreads along the interior of the axon to the next node (blue arrows), where voltage-gated sodium channels enable reinitiation. Thus, the action potential appears to jump from node to node as it travels along the axon (red arrows).



⇒ **Mastering Biology Animation: Propagation of an Action Potential in Unmyelinated and Myelinated Axons**

depolarizes the membrane and regenerates the action potential (Figure 48.15).

Action potentials propagate more rapidly in myelinated axons because the time-consuming process of opening and closing of ion channels occurs at only a limited number of positions along the axon. This mechanism for propagating action potentials is called **saltatory conduction** (from the Latin *saltare*, to leap) because the action potential appears to jump from node to node along the axon.

The major selective advantage of myelination is its space efficiency. A myelinated axon 20 μm in diameter has a conduction speed faster than that of a squid giant axon with a diameter 40 times greater. Consequently, more than 2,000 of those myelinated axons can be packed into the space occupied by just one giant axon.

For any axon, myelinated or not, the conduction of an action potential to the end of the axon sets the stage for the next step in neuronal signaling—the transfer of information to another cell. This information handoff occurs at synapses, our next topic.

CONCEPT CHECK 48.3

1. How do action potentials and graded potentials differ?
2. In multiple sclerosis (from the Greek *skleros*, hard), a person's myelin sheaths harden and deteriorate. How would this affect nervous system function?
3. How do both negative and positive feedback contribute to the changes in membrane potential during an action potential?
4. **WHAT IF?** Suppose a mutation caused gated sodium channels to remain inactivated longer after an action potential. How would this affect the frequency at which action potentials could be generated? Explain.

For suggested answers, see Appendix A.

CONCEPT 48.4

Neurons communicate with other cells at synapses

Transmission of information from neurons to other cells occurs at synapses. Synapses are either electrical or chemical.

Electrical synapses contain gap junctions (see Figure 6.30) that allow electrical current to flow directly from one neuron to another. Such synapses often play a role in synchronizing the activity of neurons that direct rapid, unvarying behaviors. For example, electrical synapses associated with the giant axons of squids and lobsters facilitate swift escapes from danger. Electrical synapses are also found in the vertebrate heart and brain.

The majority of synapses are *chemical synapses*, which rely on the release of a chemical neurotransmitter by the presynaptic neuron to transfer information to the target cell. While at rest, the presynaptic neuron synthesizes the neurotransmitter at each synaptic terminal, packaging it in multiple membrane-enclosed compartments called *synaptic vesicles*. When an action potential arrives at a chemical synapse, it depolarizes the plasma membrane at the synaptic terminal, opening voltage-gated channels that allow Ca^{2+} to diffuse in. The Ca^{2+} concentration in the terminal rises, causing synaptic vesicles to fuse with the terminal membrane and release the neurotransmitter.

Neurotransmitter released from the synaptic terminus diffuses across the *synaptic cleft*, the gap that separates the presynaptic neuron from the postsynaptic cell. Diffusion time is very short because the gap is less than 50 nm across. Upon reaching the postsynaptic membrane, the neurotransmitter binds to and activates a specific receptor in the membrane. This series of events at the synapse is summarized in Figure 48.16.